

Welcome Ethan Bandick
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Washington University in St. Louis

Category: General
Application # 150

Round: Final Report
General Comments

Judge	Score
Judge 1	Very good report. It's hard to hit all the points in the design sections of the report, but you did very well with the last 4 sections, a challenging spot for most report submissions. -A few tips for improving things, it's hard to be consistent with metric due to the DBF requirements and design tools typically available. I think if you switched to imperial, you'd have more success with staying consistent. -The drawing package was good. Consider that if the mission requires special packaging, you should look at adding drawings showing the stored configurations. Otherwise it's hard to know if you can meet those requirements.
Judge 2	
Judge 3	Excellent executive summary and very good management summary. Decomposition of mission requirements into subsystems requirements was really well done! Sensitivity study in the conceptual design section was confusing because it appeared that you held several important design parameters constant and concluded that the antenna length and payload weight should be maximized, which was obvious from the start. It seemed a few design considerations were omitted (antenna created drag, but did not appear to be addressed for S&C). Several sections would have benefitted from more careful proofreading and editing. Great detail design section! Drawing package would benefit from a few more details relevant to this year's unique mission requirements. Be careful with references, that all are included and in order in the text.
Judge 4	Executive Summary: What is the expected ground mission weight? Management Summary: How many total team members are there? Make sure to put the specific due dates on the schedule for key deliverables Need detail on schedule for subcomponent design Conceptual Design: No

Judge	Score
	mention of the coin flip to determine wings Subsystem requirements should be quantitative with expected values for wing area, thrust, etc based on mission requirements Sensitivity study seems a bit incomplete Preliminary Design: Explain software tools a bit more instead of referencing it in the bibliography Tucson is quite a bit higher than 330m It would be nice to have dimensionalized lift and drag estimates Not much discussion on static stability Good estimate on antenna yawing moment, but should explain how antenna drag was calculated Detail Design: Should include dimensions of the fuselage as well Should state max loads on landing gear in terms of weight Need to state maximum test weight for ground mission Drawing Package: Need to show interface of electronics payload with fuselage Need to show how the aircraft fits in the box Somewhat odd that the drawings don't include aircraft skin Manufacturing Plan: Need to discuss material downselect rather than state what you ended up using Need more detail on the manufacturing process Test Plan: Some sort of aero test like a wind tunnel would be nice to confirm lift and drag estimates Vague test setup descriptions Performance Results: Should use differences between predicted and test results to refine design Need to redo the flight test with the "repaired" aircraft
Judge 5	Management Summary: It would be good if the schedule included detail on subcomponent design, manufacture and testing. Conceptual Design: I'd like to see further breakdown of the mission requirements into specific, quantitative design requirements. For example, high C_{lmax} isn't a requirement. What does "high" mean? I think the sensitivity study was very well done. Preliminary Design: Why is the maximum aspect ratio 5.2? Its not clear how this number was derived. It'd be good to report how many airfoils were considered before you down selected to these 5. It's not clear to the reader that your study was exhaustive. The approach is sound otherwise. Reynolds number could be low. What if the day isn't average? It could easily be in the 90s during the competition. The plots are good but they're very hard to read. Instead of cut and pasting the plots directly output from your tools, replot this data so that it is more presentable. Stability analysis could have a little more discussion. Static stability should be plotted and discussed a little more. Also, what are the targets for your dynamic stability parameters or why is your data sufficient. You leave it up to the reader to fully interpret the data. Section 4.5 has some of the decomposed subsystem requirements I was looking for in section 3. I love that you've referenced and published your km_sim. I still think your plots are impossible to read, but overall I think this is very good. Uncertainty discussion is good. Detail Design: FEA of the wing is good but it would have been good to use it to predict ground mission performance. I don't see any ground mission analysis here as I would expect. There could have been more discussion on electronics/systems design, motor mounts, and other details. Drawing Package: It would be good to show a drawing of how everything fits in the shipping container. Test Plan: Steady state aerodynamic testing would be a good addition. I'm also surprised there isn't a wing failure test listed. The ground test is good, but I would have taken a wing to failure. Performance Results: More detail on the antenna drag measurement method (either in this section or the last section) would be a good addition. It would have been good to have some

Judge	Score
	flight testing results from your final (or proposed final) design. A propulsion endurance test was also expected.

ound: Letter of Intent/Proposal
Overall Summary

Judge	Score
Judge 1	
Judge 2	
Judge 3	

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